An Interpretable Rule-Based Medical Diagnosis Expert System for Preliminary Decision Support

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***Abstract*—This project presents an interpretable medical expert system for preliminary diagnostic decision support. The system represents user symptoms and selected medical history as facts and evaluates them against authored IF–THEN disease rules. It supports 31 diseases and provides a console interface and a Flask-based web interface. The inference process ranks rule matches using an evidence score and surfaces matched symptoms to make the reasoning traceable. Separate treatment information and PDF reporting extend the system beyond the core inference engine. The prototype is intended for educational and decision-support use rather than clinical diagnosis, since it does not include laboratory measurements, medical images, disease prevalence, or clinical validation.**

***Index Terms*—Medical expert system, rule-based reasoning, IF–THEN rules, medical diagnosis, Experta, Flask, explainable decision support.**

I. Introduction

Medical diagnosis has traditionally relied on the expertise of healthcare professionals who integrate patient history, symptoms, physical examination findings, and diagnostic tests to reach clinical conclusions. However, the growing complexity of medical knowledge and the increasing demand for healthcare services have created opportunities for computer-based decision support systems. Expert systems, which encode specialist knowledge into rule-based frameworks, offer a promising approach to providing interpretable diagnostic assistance.

This project develops an interpretable medical expert system that leverages a rule-based approach for preliminary diagnostic decision support. Unlike statistical or machine learning models that often operate as black boxes, our system stores symptoms and medical history as explicit facts and evaluates them against authored disease rules. The reasoning process is transparent, allowing users to trace how specific evidence leads to diagnostic suggestions.

The system supports 31 diseases and provides both a console interface and a web-based interface built with Flask. The inference engine, implemented using the Experta framework, evaluates rule matches and computes evidence scores to rank potential diagnoses. Treatment information and PDF report generation extend the system's utility beyond core inference.

II. Literature Review

A. Historical Evolution of Medical Expert Systems

The development of medical expert systems has a rich history dating back to the 1970s. The pioneering MYCIN system, developed by Shortliffe and colleagues at Stanford University, demonstrated the feasibility of rule-based reasoning for medical diagnosis [1], [2]. MYCIN used production rules to diagnose bacterial infections and recommend antibiotic treatments, achieving performance comparable to infectious disease experts. The system's success established rule-based reasoning as a viable approach for medical decision support.

The theoretical foundations for diagnostic reasoning were established by Ledley and Lusted [3], who introduced logical and probabilistic frameworks for medical decision-making. Their work provided the conceptual basis for systematically processing clinical data to reach diagnostic conclusions.

The INTERNIST-1 system [4] represented another significant advancement, focusing on general internal medicine. This system attempted to model the diagnostic reasoning process of expert physicians, using a large knowledge base of disease descriptions and sophisticated problem-solving strategies.

B. Rule-Based Reasoning in Medical Domains

Rule-based systems have been extensively studied and applied in medical domains due to their inherent interpretability. Buchanan and Shortliffe [5] provided comprehensive documentation of the MYCIN experiments and broader implications for building expert systems. The Rete algorithm, developed by Forgy [6], significantly improved the efficiency of rule-based systems by optimizing pattern matching, making it feasible to handle large rule bases.

Hayes-Roth, Waterman, and Lenat [7] established fundamental principles of expert systems, including knowledge acquisition, representation, and reasoning strategies. Their work highlighted the importance of maintaining transparency in expert systems, allowing users to understand the reasoning behind recommendations. Giarratano and Riley [8] provided comprehensive coverage of expert system principles and programming techniques.

C. Modern Clinical Decision Support Systems

Recent developments have focused on integrating rule-based reasoning with more sophisticated approaches. Sutton *et al.* [9] provided an overview of clinical decision support systems, discussing benefits, risks, and strategies for success. Their analysis emphasized the importance of integrating CDS systems into clinical workflows while maintaining user engagement and trust.

Mustafa, Saad, and Rizkallah [10] explored combining case-based reasoning and rule-based systems for medical diagnosis, demonstrating that hybrid approaches can leverage the strengths of multiple reasoning strategies. The World Health Organization [11] has published guidelines recommending digital interventions for health system strengthening, and the Agency for Healthcare Research and Quality [12] maintains parallel guidance on clinical decision support.

D. Explainability and Trust in Medical AI

The importance of explainability in medical AI has been increasingly recognized. Rudin [13] argued strongly for using interpretable models rather than black-box approaches in high-stakes decisions, particularly in healthcare. Her work highlighted that interpretable models, such as rule-based systems, can provide the transparency needed for clinical decision-making.

Amann *et al.* [14] provided a multidisciplinary perspective on explainability in healthcare AI, discussing technical, legal, ethical, and clinical considerations. Their analysis emphasized that explainability must serve various stakeholders, including clinicians, patients, and regulators.

Shortliffe and Sepúlveda [15] discussed the evolution of clinical decision support in the era of AI, emphasizing the continued relevance of rule-based approaches alongside newer machine learning methods [16].

E. Technical Frameworks and Implementation

Modern rule-based systems benefit from frameworks like Experta [18], which provides a Python-based implementation of the Rete algorithm. The Experta framework enables efficient rule-based reasoning while maintaining transparency. Web-based implementations using frameworks like Flask [19] have made it possible to deploy medical expert systems as accessible web applications.

F. Challenges and Future Directions

Despite their advantages, rule-based systems face several challenges. Knowledge acquisition remains significant, requiring expert involvement in rule development and maintenance. Rule-based systems also face difficulties with uncertainty management and cannot easily adapt to new patterns without explicit rule updates [17]. Future directions include integrating rule-based systems with machine learning approaches, developing more sophisticated uncertainty handling, and creating systems that can learn while maintaining transparency.

III. System Design

The system architecture follows a component-based design where user input flows through multiple layers. First, user input enters through the console layer or from the web. Then, a response manager organizes inputs and converts them into structured facts. The inference core processes these facts and evaluates them against disease rules. Finally, disease knowledge is stored in separate HTML and Markdown files, allowing treatment information and medical advice to be updated independently. Figs. 1 and 2 show the component flow and the inference workflow respectively.

A. Knowledge Representation

The knowledge base organizes symptoms and findings into distinct categories to support systematic reasoning. General symptoms like fever, fatigue, and weakness provide broad evidence. Respiratory findings including cough, breathing difficulty, and chest pain support respiratory disease rules. Gastrointestinal findings such as vomiting, abdominal pain, and acidity contribute to digestive disease rules. Eye-related findings like redness, itching, and watery discharge support eye-related rules.

The evidence score for a disease d is computed as:

$$\text{Score}(d) = \frac{\text{Number of matched symptoms for disease } d}{\text{Total number of required symptoms for } d} \tag{1}$$

This score is not treated as a clinical diagnosis but as an indicator of evidence supporting the particular disease.

B. Inference Process

The rule set is designed to avoid treating a generalized symptom as sufficient by itself. For example, fever is a common symptom in multiple diseases like dengue, malaria, influenza, or pneumonia. Therefore, the system combines multiple facts and findings before suggesting a possible diagnosis. Taking the dengue rule as an example, high fever, body pain, rash, and bleeding tendency are required together to reach a conclusion.

IV. Implementation

A. Technology Stack

The system is implemented using Python as the primary programming language. The Experta framework provides rule-based expert-system and inference logic. Flask serves as the local web application framework for the browser interface. HTML and CSS are used for the browser interface, while Markdown and HTML files store separate disease-information content. PDF generation enables saving consultation results as reports.

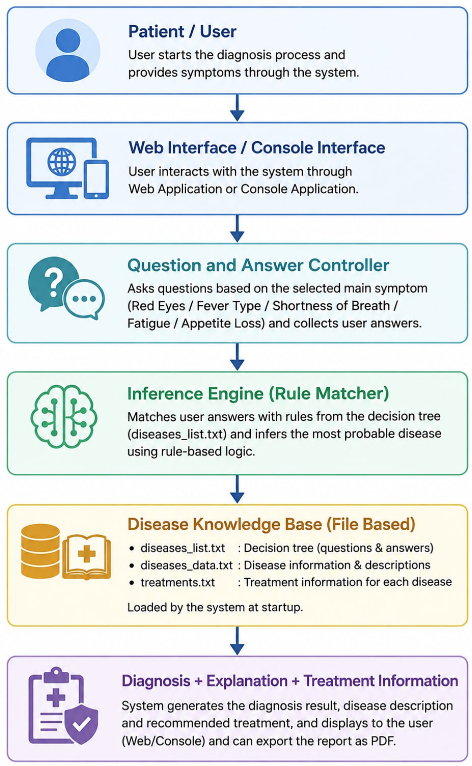


Fig. 1. Component flow from consultation to explainable output.

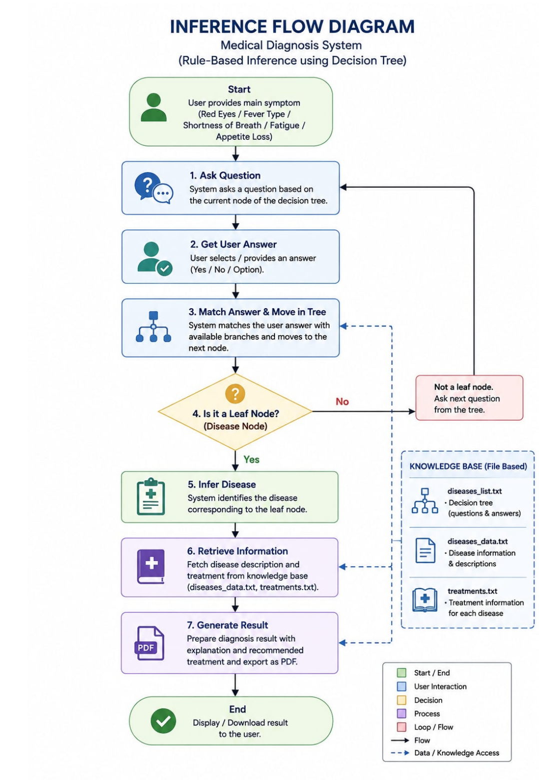


Fig. 2. Inference workflow with explicit evidence construction and rule evaluation.

B. Console Engine

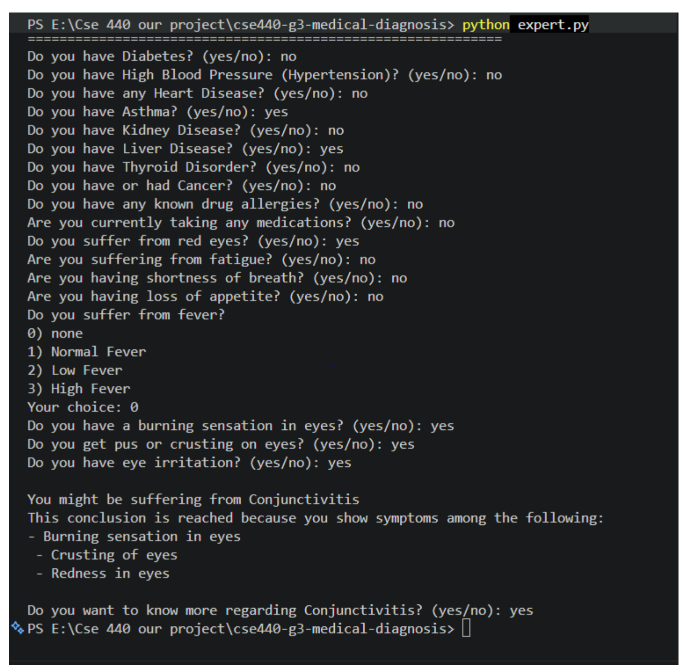
The console engine, implemented in `expert.py`, operates independently of the web interface. It asks a series of questions, records responses, uses the rule engine, and prints possible diagnoses in the terminal (Fig. 3). This implementation helps users understand the expert system core and serves those who prefer a command-line interface.

The consultation process follows distinct stages:

- **Input:** Records responses and provides follow-up questions
- **Session:** Associates responses with the running consultation session
- **Reasoning:** Rule engine looks for disease conditions based on collected data
- **Result:** Identifies possible conditions and displays matched evidence
- **Follow-up:** Finalizes disease and makes downloadable PDF accessible

C. Web Application

The web application, implemented in `web.py` using Flask, transforms the console-like response into a step-by-step question interaction (Figs. 4 and 5). Users can view and respond by clicking, making the system more accessible. Information is stored only for the duration of the session, leading to user reports and PDF output.


```
PS E:\Cse 440 our project\cse440-g3-medical-diagnosis> python expert.py
============================================================
Do you have Diabetes? (yes/no): no
Do you have High Blood Pressure (Hypertension)? (yes/no): no
Do you have any Heart Disease? (yes/no): no
Do you have Asthma? (yes/no): yes
Do you have Kidney Disease? (yes/no): no
Do you have Liver Disease? (yes/no): yes
Do you have Thyroid Disorder? (yes/no): no
Do you have or had Cancer? (yes/no): no
Do you have any known drug allergies? (yes/no): no
Are you currently taking any medications? (yes/no): no
Do you suffer from red eyes? (yes/no): yes
Are you suffering from fatigue? (yes/no): no
Are you having shortness of breath? (yes/no): no
Are you having loss of appetite? (yes/no): no
Do you suffer from fever?
0) none
1) Normal Fever
2) Low Fever
3) High Fever
Your choice: 0
Do you have a burning sensation in eyes? (yes/no): yes
Do you get pus or crusting on eyes? (yes/no): yes
Do you have eye irritation? (yes/no): yes

You might be suffering from Conjunctivitis
This conclusion is reached because you show symptoms among the following:
- Burning sensation in eyes
 - Crusting of eyes
 - Redness in eyes

Do you want to know more regarding Conjunctivitis? (yes/no): yes
PS E:\Cse 440 our project\cse440-g3-medical-diagnosis>
```

Fig. 3. Console log of `expert.py` showing the consultation process.

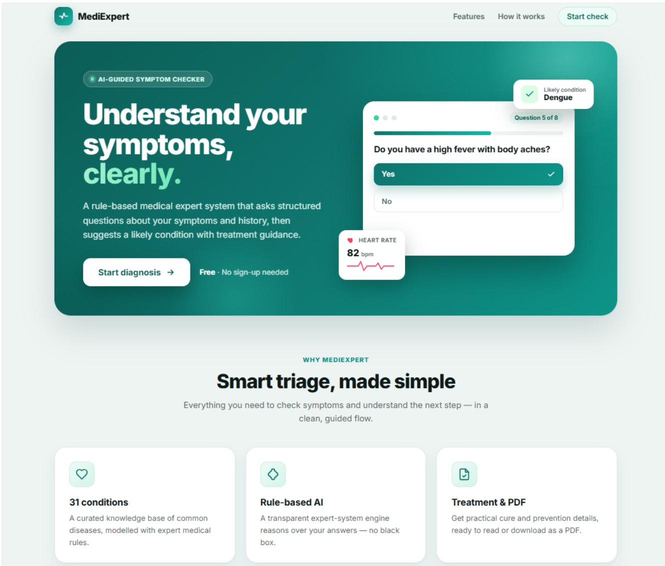


Fig. 4. Main landing page of the web interface.

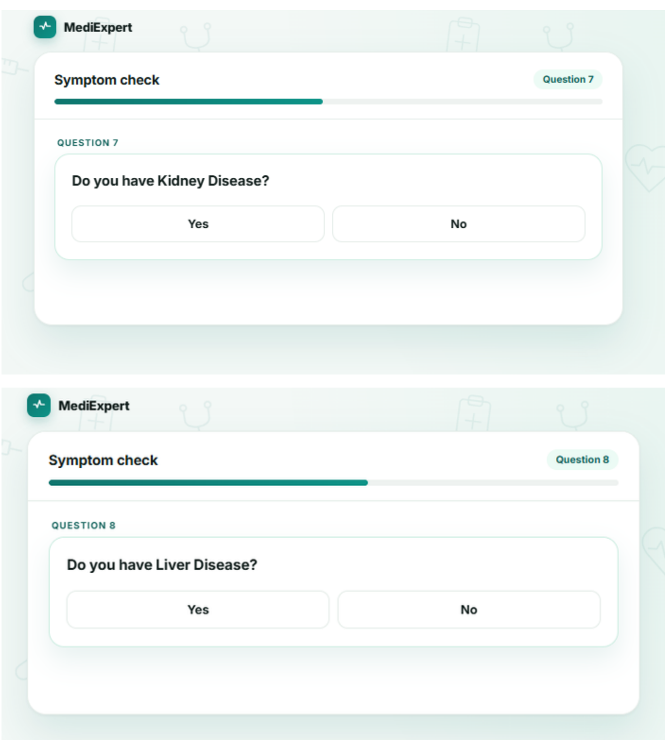


Fig. 5. Sample questions asked by the system during a consultation.

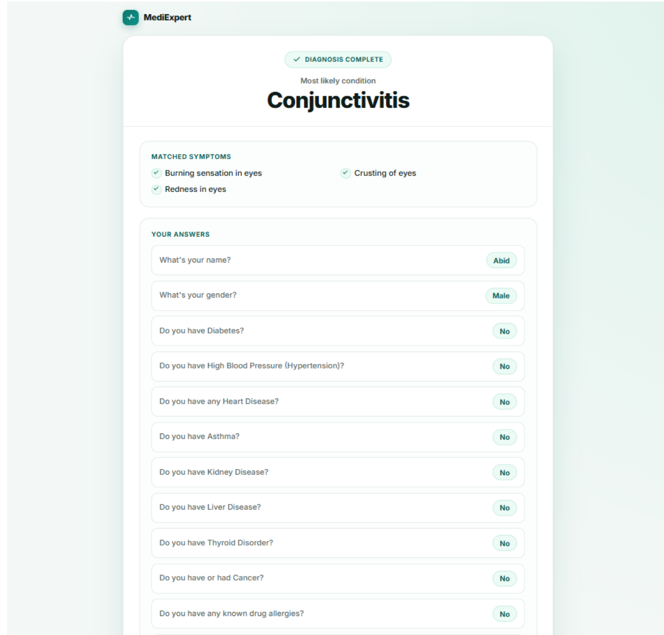


Fig. 6. Detected disease shown after all questions are answered, with the matched symptoms listed as supporting evidence.

When the consultation completes, the result page names the most likely condition together with the matched symptoms that support it, and reproduces the full answer trace so the user can review every response (Figs. 6 and 7).

D. *Treatment Content and Reporting*

Selecting the treatment option from the result page opens the treatment reference for the diagnosed condition (Fig. 9). This content is authored in separate Markdown and HTML files outside the rule engine, so disease information can be revised without touching the diagnostic logic. The same consultation may also be exported as a PDF report containing the patient summary, the predicted condition, the matched symptoms, and the treatment guidance (Fig. 8).

V. Disease Knowledge Base

The knowledge base encompasses 31 conditions across multiple categories, including metabolic, cardiovascular, respiratory, gastrointestinal, infectious, vector-borne, and eye-related diseases. Table I lists every supported condition.

A. *Distribution by Category*

Fig. 10 shows how the 31 conditions distribute across the five disease categories. Fever-driven conditions form by far the largest group at sixteen, which reflects both their prevalence and the fact that they are the group most in need of rule-based discrimination: these conditions share a non-specific febrile presentation and cannot be separated by any single finding.

Table II lists the symptom domains used to organise the questions and the findings that are representative of each.

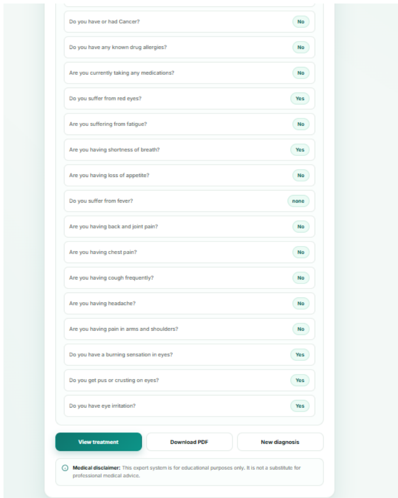


Fig. 7. Full answer trace with the available follow-up actions and the educational-use disclaimer.

Fig. 8. Generated PDF report as it is saved when the user selects the download option.

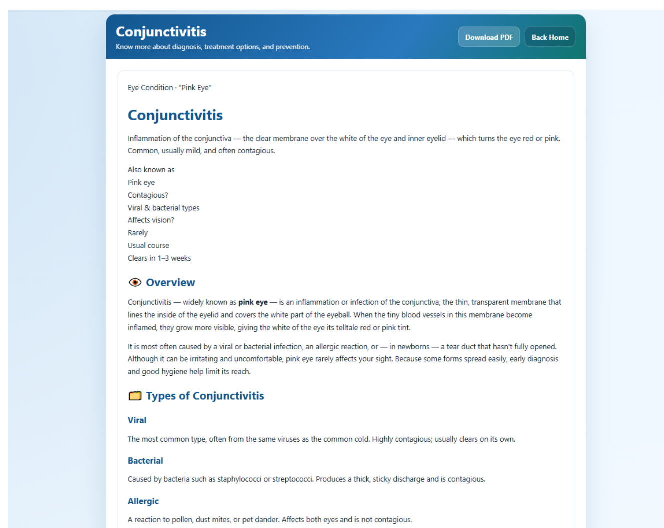


Fig. 9. Treatment reference page opened from the result screen.

TABLE I
SUPPORTED DISEASES

1.	Diabetes	17.	Coronary arteriosclerosis
2.	Pneumonia	18.	Asthma
3.	Anemia	19.	Hypothyroidism
4.	AIDS/HIV infection	20.	Dehydration
5.	Obesity	21.	Arthritis
6.	Bronchitis	22.	Pancreatitis
7.	Peptic ulcer	23.	Gastritis
8.	Hepatitis	24.	Influenza
9.	Tuberculosis	25.	Malaria
10.	Dengue	26.	Coronavirus disease
11.	Conjunctivitis	27.	Eye allergy
12.	Typhoid	28.	Chikungunya
13.	Sinusitis	29.	Common cold
14.	UTI	30.	Gastroenteritis
15.	COPD	31.	Hyperthyroidism
16.	Dry eye syndrome		

TABLE II
DISEASE CATEGORIES AND REPRESENTATIVE SYMPTOMS

Domain	Representative symptoms
Respiratory	Cough, chest pain, breathing difficulty
Gastrointestinal	Vomiting, abdominal pain, acidity
Vector-borne	Fever, chills, rash, body pain
Metabolic	Thirst, frequent urination, weight change
Eye-related	Redness, itching, watery discharge
General	Fatigue, weakness, fever

B. *Representative Rules*

The system implements representative rules for each disease category:

- **Respiratory:** high fever + breathing difficulty → possible pneumonia
- **Metabolic:** frequent urination + excessive thirst + weight loss → possible diabetes
- **Eye related:** redness + itching + watery discharge → possible eye allergy

VI. TESTING AND EVALUATION

Testing focused on whether expected symptom combinations from consultations activate the intended rules, and on

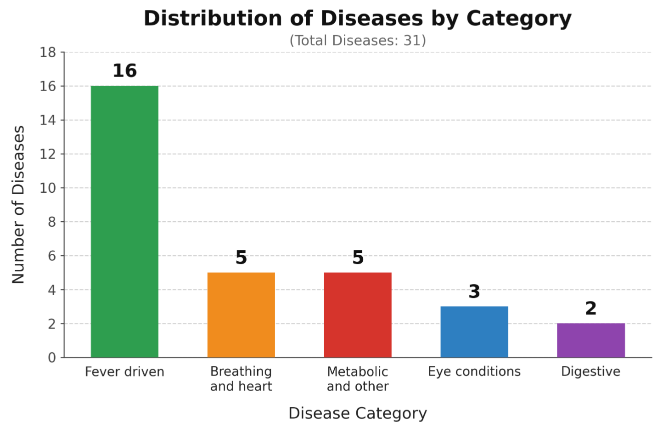


Fig. 10. Distribution of the 31 supported conditions across disease categories. Counts are derived from the question branch each condition belongs to.

TABLE III
TEST SCENARIOS AND EXPECTED OUTCOMES

Scenario	Input pattern	Expected outcome
Pneumonia	Fever + cough + breathing difficulty	Pneumonia suggested
Dengue	High fever + body pain + rash	Dengue suggested
Diabetes	Thirst + frequent urination + weight loss	Diabetes suggested
Eye allergy	Itching + redness + watery discharge	Eye allergy suggested
Generic fever	Fever only	No overconfident diagnosis
Incomplete input	Required answer missing	Consultation continues
Treatment lookup	Known disease	Correct treatment page

the system's behaviour when insufficient evidence exists for a confident conclusion. Table III lists the scenarios and their expected outcomes.

A. Example Consultation

Table IV follows a single consultation from the first reported symptom to the explained conclusion.

TABLE IV
EXAMPLE CONSULTATION FLOW

Stage	Interaction
Initial symptom	User confirms fever and specifies it is high
Respiratory evidence	User reports coughing and difficulty breathing
Additional evidence	Chest pain also reported
Rule outcome	Collected findings support possible pneumonia
Explanation	System identifies high fever, cough, breathing difficulty and chest pain as the supporting evidence

B. What the Tests Do Not Establish

The project ensures the implemented logic behaves consistently with the tested patterns, but it does not claim clinical accuracy. Disease identification requires physical diagnosis and clinical tests reviewed by medical experts.

VII. DISCUSSION AND LIMITATIONS

A. Strengths

The system's primary strengths include visible reasoning and decision-making, directly modifiable rules, multiple interface options (web and console), and a clear implementation of an expert system architecture.

B. Constraints

The knowledge rules are manually authored and do not include laboratory measurements, medical images, disease prevalence, or probability modelling. The software cannot substitute for physical examination or professional diagnosis. Different diseases can share similar symptoms, reducing certainty, and the knowledge base would require expert review before any real clinical deployment.

C. Safety and Ethical Use

Diagnostic suggestions can create either false reassurance or unnecessary concern if interpreted as confirmed medical findings. The system should be presented explicitly as an educational and decision-support tool. Emergency symptoms including severe chest pain, major breathing difficulty, significant bleeding, and severe dehydration should be handled by qualified healthcare professionals.

D. Future Development

Future development plans include moving rules into a database or JSON-based store with source recording and review dates, adding severity assessment and red-flag warnings, providing a clinician review mode, expanding the disease and symptom library, improving the web consultation experience, introducing multilingual support, automating unit tests for individual disease rules, and adding probabilistic ranking when several conditions match.

VIII. CONCLUSION

This project demonstrates how logic programming and rule-based knowledge representation can be combined to create an interpretable medical diagnosis support prototype. User symptoms and selected history are transformed into structured facts, disease rules are evaluated against those facts, and the system returns possible conditions with supporting evidence. The implementation includes both a console engine and a Flask web application, while separate treatment content and PDF reporting extend the system beyond the core inference process.

The key contribution is explainability: reasoning can be followed from user-provided evidence to matched rules rather than being hidden inside an opaque prediction model. The absence of clinical validation means the prototype should remain within an educational and decision-support context.

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